

# Frédérique LECOURTIER

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📧 (private)

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*PhD student*

(Last updated on May 19, 2026)

## Research interests

Scientific computing • Physics-informed machine learning • Numerical analysis • Finite element methods • Hybrid solvers • Partial differential equations • Error estimation

## Education

2023 - ... **PhD** in Applied Mathematics, *University of Strasbourg*, France

*Team:* Mimesis, Inria Center at the University of Lorraine.

*Supervised by:* Emmanuel Franck, Michel Duprez and Vanessa Lleras.

2021 - 2023 **Master** in Scientific Computing and Mathematics of Information (CSMI), *University of Strasbourg*, France - Honors

2018 - 2021 **Bachelor** in Applied Mathematics, *University of Strasbourg*, France - Honors

2018 **Scientific Baccalaureate**, *Lycée LeCorbusier*, Illkirch, France

## Experience

2023 - ... **PhD** in Applied Mathematics, *Mimesis*, Strasbourg, France

Python PyTorch Jax FEniCSx ScimBa

*Subject:* Finite element methods and neural networks for parametric PDEs

*Objectives:*

- Develop hybrid methods combining finite element methods and neural networks to solve parametric PDEs.
- Develop scientific machine learning methods in ScimBa (a Python library under development in the Macaron Inria team) for solving parametric PDEs.

🔗 [GitHub repository.](#)

2023 **Internship**, *Mimesis*, Strasbourg, France

(6 months)

Python Tensorflow PyTorch FEniCS GitHub VSCode

*Subject:* Finite element methods and neural networks for augmented surgery.

*Supervised by:* Emmanuel Franck, Michel Duprez and Vanessa Lleras.

🔗 [GitHub repository.](#) 🔗 [Report.](#)

2022 **Internship**, *Cemosis*, Strasbourg, France

(2 months)

C++ Python MPI Feel++ Antora Sphinx Doxygen VSCode CMake

*Subject:* Parareal method for solving the heat equation in Feel++.

*Supervised by:* Christophe Prudhomme and Luca Berti.

*Objectives:*

- Understand and implement the Parareal method in Python and C++, a parallel-in-time integration method.
- Implement the heat equation solver in Feel++ (a C++ library developed by the Cemosis team, IRMA).

🔗 [GitHub repository.](#) 🔗 [Report.](#)

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## Publications

- 2026 H. Barucq, M. Duprez, F. Faucher, E. Franck, [F. Lecourtier](#), V. Lleras, V. Michel-Dansac, N. Victorion. **Enriching continuous Lagrange finite element approximation spaces using neural networks**. Mathematical Modelling and Numerical Analysis (M2AN). ([pdf](#)), ([preprint \[arXiv, HAL\]](#))

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## Conferences, presentations and posters

- 2026 **SciCADE 2026**, *George Square campus*, Edinburgh, United Kingdom  
29 June-3 July [Website](#).
- 2026 **CANUM 2026**, Saint-Jacut-de-la-Mer, France  
1-5 June [Website](#).
- 2025 **PhD seminar**, *IMAG Institute*, Montpellier, France  
17 December [Website](#). [Presentation](#).
- 2025 **ICOSAHOM 2025**, *McGill University*, Montreal, Canada  
13-18 July [Website](#). [Presentation](#).
- 2025 **EMS-TAG-SciML-25**, *Polytechnic University of Milan*, Milan, Italy  
24-26 March [Website](#). [Poster](#).
- 2025 **DTE & AICOMAS 2025**, *Arts et Metiers - ENSAM*, Paris, France  
17-21 February [Website](#). [Presentation](#).
- 2024 **CJC-MA 2024**, *Ecole Normale Supérieure de Lyon*, Lyon, France  
28-30 October [Website](#). [Poster](#).
- 2024 **SciML 2024**, *UFR de mathématique et d'informatique*, Strasbourg, France  
8-12 July [Website](#). [Poster](#).

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## Teaching experience

- 2025 - 2026 **Teaching assignment**, *UFR de mathématique et d'informatique*, Strasbourg, France  
50 h of computer science labs - L1 (Python) / L2 (Python).  
14 h of integrated lectures in computer science - L2 (Python).
- 2025 **Rendez-vous des Jeunes Mathématiciennes et Informaticiennes (RJMI)**, *UFR de mathématique et d'informatique*, Strasbourg, France  
(3-day event) Supervision of research workshops for a group of high school students as part of the RJMI event.
- 2024 - 2025 **Teaching assignment**, *UFR de mathématique et d'informatique*, Strasbourg, France  
64 h of computer science labs - L1 (Python) / L2 (Python) / L3 (C++).
- 2023 - 2024 **Teaching assignment**, *UFR de mathématique et d'informatique*, Strasbourg, France  
50 h of computer science labs - L2 (Python) / L3 (C++).

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## Skills

GitHub · Python · C++ · C · Rust · LaTeX · PyTorch · JAX · ParaView · Gmsh · FEniCS

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## References

### PhD advisor:

Emmanuel Franck : [emmanuel.franck@inria.fr](mailto:emmanuel.franck@inria.fr)

### PhD co-supervisors:

Michel Duprez : [michel.duprez@inria.fr](mailto:michel.duprez@inria.fr)

Vanessa Lleras : [vanessa.lleras@umontpellier.fr](mailto:vanessa.lleras@umontpellier.fr)

### Collaborator:

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